Engineering Encyclopedia

A Comprehensive Reference for Modern Engineers

What is the Engineering Encyclopedia?

The JC Tech Engineering Encyclopedia is a curated, ever-growing digital reference that spans the core disciplines of modern engineering. From fundamental mechanical principles to cutting-edge AI-driven manufacturing, this encyclopedia is designed to be the single source of truth for practicing engineers, students, and technical decision-makers.

Core Disciplines Covered

- Mechanical Engineering: Thermodynamics, fluid mechanics, machine design, vibration analysis, and material science.
- Electrical & Electronics: Circuit theory, power systems, embedded systems, signal processing, and control theory.
- Manufacturing & Production: CNC machining, additive manufacturing, casting, forging, welding, and quality control (Six Sigma, SPC).
- Automotive & EV Engineering: Powertrain design, battery technology, motor controllers, regenerative braking, and thermal management.
- Computer-Aided Engineering: FEA, CFD, topology optimization, CAD/CAM workflows, and simulation-driven design.
- AI & Automation in Engineering: Machine learning for predictive maintenance, generative design, robotic process automation, and digital twins.

How to Use This Resource

Each topic in the encyclopedia includes a concise definition, mathematical formulations where applicable, real-world application examples, and references to industry standards (ISO, ASME, SAE, BIS). Engineers can

search by keyword, browse by discipline, or follow curated learning paths.

Featured Topics (Sample Entries)

Stress-Strain Curve:	Relationship between applied force and material deformation; elastic vs. plastic region.
Reynolds Number:	Dimensionless ratio predicting laminar vs. turbulent flow (Re = rho * v * D / mu).
GD&T (Geometric Dimensioning):	ASME Y14.5 standard for defining tolerances on engineering drawings.
Regenerative Braking:	Energy recovery system converting kinetic energy back to electrical storage in EVs.

Why JC Tech Built This

Most engineering references online are fragmented, outdated, or hidden behind paywalls. We built the Engineering Encyclopedia to democratize technical knowledge. Whether you are a fresh graduate preparing for interviews or a senior engineer refreshing a concept before a design review, this resource saves you time and ensures accuracy.

Key Statistics

- 500+ engineering terms and definitions curated by industry experts.
- Covers 6 major engineering disciplines with cross-references.
- Updated quarterly with new topics and industry trends.
- Used by 10,000+ engineers and students worldwide.

Access & Updates

The Engineering Encyclopedia is freely available through the JC Tech Knowledge Hub. Subscribe to our newsletter for quarterly updates, or contact us for custom enterprise editions tailored to your organization's technical domains.

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